Book I

Primitive Relational Foundations

Elements of Nested Fibrational Cosmology

Contents

I.0 Purpose

Book I fixes the primitive admissible arena of the canonical spine. It begins from a purely finite relational setting and develops only the minimal theorem-bearing grammar needed for admissible tests, observational equivalence, quotient structure, refinement, finite stabilization, and primitive defect bookkeeping.

No geometric, continuum, probabilistic, or branch-specific structure is primitive here. Such language, if it later appears in the spine, must be licensed by later books. Book I therefore establishes the first governing rule of the program:

What has not yet been derived is not yet licensed.

Book I does *not* prove locality stability, source universality, branch legitimacy, continuum correspondence, or any physical interpretation. It proves only the primitive theorem-bearing kernel: admissible tests, observational quotient, finite refinement stabilization, and defect bookkeeping.

1. Primitive Stance

We work in a purely finite relational setting throughout this book. Observable content is generated only by admissible internal tests; token-level identity has no canonical force beyond membership in a finite site set.

Standing Rule 1.1 ([D]). No primitive spacetime, metric, probability space, or background field is assumed.

Standing Rule 1.2 ([D]). A structure may appear in a theorem only when its generating conditions have been derived without presupposition.

Standing Rule 1.3 ([D]). No observable distinction may be drawn by appealing to external naming, primitive coordinates, or hidden memory of raw labels.

Remark 1.4 ([R]). Standing Rules 1.1–1.3 are not denials that geometric, continuum, or probabilistic language may later become lawful. They are the claim that none of it is primitive at the base of the spine. Such language must be *earned* through certified transfer; it cannot be smuggled into the primitive grammar.

The following six conditions are the *framework-collapse gates*: structural preconditions whose violation makes admissible observational comparison impossible. Any finite relational system failing one of these gates collapses: no stable quotient, no well-defined defect ledger, and no downstream canonical force.

Standing Rule 1.5 ([D]). (CG-1: Finite regime alphabet.) The set of stabilized collar-type classes is finite. An infinite collar-type alphabet prevents stabilization of the observational quotient and collapses the framework.

Standing Rule 1.6 ([D]). (CG-2: Finite fibers.) Every equivalence class in the observational quotient is finite. Infinite fibers destroy the defect arithmetic and collapse the bookkeeping.

STANDING RULE 1.7 ([D]). (CG-3: Terminating projection.) The quotient projection $\pi : \Omega \to \Sigma_n$ has no non-terminating reduction chain. Non-termination collapses the stabilization argument.

STANDING RULE 1.8 ([D]). (CG-4: No identity smuggling.) No admissible test may produce a canonical inverse to the quotient map at a singular interface. Identity smuggling at singular points re-introduces presentation-level surplus into the canonical grammar.

STANDING RULE 1.9 ([D]). (CG-5: Finite nesting depth.) The fibrational nesting depth of any licensed configuration is bounded. Infinite nesting prevents the finite stabilization argument and collapses the spine.

STANDING RULE 1.10 ([D]). (CG-6: Finite valence.) Every site in every licensed configuration has finitely many relational neighbours. Infinite valence collapses the local observational structure on which the defect bookkeeping depends.

REMARK 1.11 ([R]). Standing Rules 1.5–1.10 are not independently postulated constraints; they follow from the requirement that the framework is persistence-selectable. Any system violating one of these gates cannot sustain the admissible quotient construction and exits the canonical program. They are recorded here at the primitive level so that later books may invoke them by name without returning to the finiteness arguments.

2. Finite Relational Arena

DEFINITION 2.1 ([D]). A *finite relational signature* is a finite family

$$L = \{R_i\}_{i \in I}$$

in which each R_i has finite arity $k_i \in \mathbb{N}$.

DEFINITION 2.2 ([D]). A *finite configuration* is a finite L-structure

$$\omega = \big(X,\ \{R_i^\omega\}_{i \in I}\big),$$

where X is a finite set of *sites* and $R_i^\omega \subseteq X^{k_i}$ for each $i \in I$.

DEFINITION 2.3 ([D]). A *configuration space* is a finite set Ω of finite configurations over the same base signature L.

DEFINITION 2.4 ([D]). Two configurations $\omega, \omega' \in \Omega$ are *structurally equivalent*, written $\omega \cong \omega'$, if there exists an L-isomorphism between them.

REMARK 2.5 ([R]). Structural equivalence compares raw relational presentations. It does not yet determine what admissible testing can distinguish. Book I is not interested in presentation richness by itself; it is interested in what survives admissible observability. The objects that carry canonical force are observational equivalence classes, not raw isomorphism types.

3. Admissible Processes, Tests, and Observational Equivalence

DEFINITION 3.1 ([D]). An *extension process* is a map

$$E : \Omega \to \mathrm{Str}^{\mathrm{fin}}_{L_E},$$

where $L_E \supseteq L$ is a finite expansion of the base signature.

DEFINITION 3.2 ([D]). A class $\mathcal{E}$ of extension processes is *admissible* if it satisfies the following four conditions.

(A1) *Closure under typed composition.* Whenever the composition of two processes in $\mathcal{E}$ is defined, the composite is also in $\mathcal{E}$.
(A2) *Structural invariance.* $\omega \cong \omega'$ implies $E(\omega) \cong E(\omega')$ for all $E \in \mathcal{E}$.
(A3) *Base-signature accountability.* Observational distinctions induced by E remain readable at the base-signature level and do not smuggle external labels.
(A4) *Finite output control.* For every $E \in \mathcal{E}$, the image $E(\Omega)$ has finitely many isomorphism types.

REMARK 3.3 ([R]). Axiom (A3) is the anti-smuggling gate at the process level. A process that produces external labels or hidden coordinates violates this axiom and is not admissible. The formal identity extension may serve as a composition unit, but it is not automatically a licensed observational test; permitting it as such would allow primitive re-identification at the ground floor.

REMARK 3.4 ([R]). (Architectural note: v6 condition (iii).) An earlier form of this framework included a condition "No New Invariant Partitions": any partition invariant under all extensions must be determined by the base incidence structure. In the canonical formulation this is split into Theorem 3.17 (maximality as theorem, not axiom) and Axiom (A3) (anti-smuggling content).

REMARK 3.5 ([R]). (Architectural note: v6 condition (iii).) An earlier form of this framework included a condition "No New Invariant Partitions": any partition of Ω invariant under all extensions in $\mathcal{E}$ must be determined by the base incidence structure. In the canonical formulation this condition is split: its maximality content is captured by Theorem 3.17 (a theorem, not an axiom), and its anti-smuggling content by Axiom (A3). The canonical axiom set A1–A4 is therefore strictly cleaner.

DEFINITION 3.6 ([D]). A *licensed internal test family* at stage n is a finite family $\mathcal{T}_n$ of admissible relational processes whose outputs are evaluated only through structures already licensed at stage n. Tests in $\mathcal{T}_n$ may not appeal to external naming, primitive coordinates, or hidden memory of raw labels.

DEFINITION 3.7 ([D]). For $\omega, \omega' \in \Omega$, write $\omega \sim_n \omega'$ if for every test $T \in \mathcal{T}_n$, the outputs $T(\omega)$ and $T(\omega')$ yield the same currently licensed observable outcome, or are both undefined. We call $\sim_n$ *observational equivalence at stage n.*

LEMMA 3.8 ([U]). *Observational equivalence $\sim_n$ is an equivalence relation on Ω.*

PROOF. *Reflexivity.* Every test $T \in \mathcal{T}_n$ returns the same outcome on ω as on itself.
Symmetry. The condition is symmetric in ω and ω' by definition.

Transitivity. If no test in $\mathcal{T}_n$ separates ω from ω', and no test separates ω' from ω'', then no test separates ω from ω''. □

DEFINITION 3.9 ([D]). The *observational quotient at stage n* is

$$\Sigma_n \ := \ \Omega/{\sim_n}.$$

DEFINITION 3.10 ([D]). A *projection interface* is a partial reduction map $\pi : \Omega \dashrightarrow \Omega_{\mathrm{red}}$ that preserves currently licensed observable behavior while forgetting finer relational structure.

DEFINITION 3.11 ([D]). A projection interface π is *singular at $y \in \Omega_{\mathrm{red}}$* if there exist distinct $\omega_1, \omega_2 \in \Omega$ with $\pi(\omega_1) = \pi(\omega_2) = y$, and no licensed inverse selection rule at y distinguishes ω_1 from ω_2.

PROPOSITION 3.12 ([U]). *If there exist $\omega_1 \neq \omega_2$ in Ω with $\omega_1 \sim_n \omega_2$, then the canonical quotient map $q : \Omega \to \Sigma_n$ is singular at the class $[\omega_1]_{\sim_n} = [\omega_2]_{\sim_n}$.*

PROOF. Because $\omega_1 \sim_n \omega_2$, both map to the same quotient class. Because $\omega_1 \neq \omega_2$, the map is many-to-one at that class. Any admissible inverse selector would distinguish ω_1 from ω_2, contradicting observational equivalence. □

REMARK 3.13 ([R]). Proposition 3.12 records an inescapable structural feature: wherever the observational quotient map is many-to-one, canonical inverse selection is impossible without illicit discrimination. The absence of a canonical inverse is not a defect of formalization; it is a genuine structural fact that later books must respect. All certified downstream structure must factor through the quotient, not through presentation-level surplus.

COROLLARY 3.14 ([U]). *The observational quotient Σ_n is well-defined, and canonical observational content at stage n is determined by admissible test equivalence, not by raw token identity or structural equivalence.*

PROOF. Lemma 3.8 ensures $\sim_n$ is an equivalence relation, so $\Omega/{\sim_n}$ is well-defined. The second clause follows from the definition of $\sim_n$: the only distinctions that survive to Σ_n are those that some licensed test in $\mathcal{T}_n$ can detect. □

LEMMA 3.15 ([U]). (Behavioral Congruence Canonicity.) *Among all equivalence relations on Ω induced by test families $\mathcal{T}'$ satisfying the admissibility constraints* (A1)–(A4)*, the observational equivalence $\sim_n$ is the finest: if $\omega \sim_n \omega'$ then $\omega \approx \omega'$ for every such $\approx$.*

PROOF. Any test family $\mathcal{T}'$ satisfying (A1)–(A4) is a subset of the full admissible family $\mathcal{T}_n$. If ω and ω' are indistinguishable by every test in $\mathcal{T}_n$, they are in particular indistinguishable by every test in $\mathcal{T}' \subseteq \mathcal{T}_n$. Hence $\omega \sim_n \omega'$ implies $\omega \approx \omega'$ for every admissibility-consistent $\approx$. □

REMARK 3.16 ([R]). Lemma 3.15 confirms that the observational quotient Σ_n is not an arbitrary choice: it is the canonical finest partition consistent with the admissibility axioms. No admissibility-consistent equivalence relation can distinguish more than $\sim_n$ distinguishes. This is the operational force of the no-smuggling discipline at the level of equivalence structure.

THEOREM 3.17 ([U]). (Admissible-Class Canonicity.) *Let (Ω, L) be a finite relational substrate with finite compression space $|\Sigma_\infty| < \infty$. Define*

$$E_{\max} := \Big\{ E : \Omega \to \mathrm{Str}^{\mathrm{fin}} \;\Big|\; E \textit{ satisfies (A1)–(A4) and } E(\Omega) \textit{ has fin. many isom. types} \Big\}.$$

Then $E_{\max}$ is the unique maximal admissible class: every admissible class $\mathcal{E}$ satisfies $\mathcal{E} \subseteq E_{\max}$.

PROOF. By definition, $E_{\max}$ consists of every map satisfying (A1)–(A4) together with finite output control. Any admissible class $\mathcal{E}$ satisfies (A1)–(A4) by definition, so $\mathcal{E} \subseteq E_{\max}$. Uniqueness is immediate since $E_{\max}$ is defined as the set of all such maps; any two descriptions of the maximal admissible class must coincide. □

REMARK 3.18 ([R]). Theorem 3.17 shows that the admissibility axioms determine a canonical ceiling on what counts as an admissible extension. Every concrete admissible class studied in the program is a sub-class of $E_{\max}$. No admissible test can exceed the bounds set by (A1)–(A4) without exiting the canonical framework.

STANDING RULE 3.19 ([D]). (Collar alphabet size K_0.) Let $K_0 \in \mathbb{N}$ denote the number of distinct stabilized WL-1 collar-type classes of the canonical finite relational substrate under the full admissible class $E_{\max}$. This number is a structural invariant of the substrate, determined by the stabilized quotient Σ_∞.

THEOREM 3.20 ([U]). (Primality and Uniqueness of the Collar Alphabet.) *The collar alphabet size K_0 is prime. Moreover, $K_0 = 7$ is the unique prime satisfying the structural self-consistency condition $F(p) = p$, where $F(p) := |\Phi(\mathfrak{g}_{\mathrm{color}}(p))| + 1$.*

PROOF. *Primality.* By Axiom (A2) (structural invariance), the WL-1 refinement assigns the same color to any two collar symbols in the same $\mathrm{Aut}(\mathbb{Z}_n)$-orbit. For composite n, proper non-trivial orbits exist, so the number of WL-1 colors is at most $n + 1 - \varphi(n) \le n - 1 < n$; hence $K_0 < n$ for composite n. For prime p, the generator $\mathtt{swap}_u \in \mathcal{E}_0$ acts as $+1 \pmod p$; since $\mathbb{Z}_p$ has no proper subgroups, the WL-1 walk visits all p symbols, giving $K_0 = p$.

Uniqueness at $K_0 = 7$. The interface coupling constraint selects for each odd prime p a canonical partition $p = a_1 + a_2 + \cdots$; the resulting gauge algebra $\mathfrak{g}_{\mathrm{color}}(p)$ satisfies the Weyl structure identity $p = |\hat{\Phi}(\mathfrak{g}_{\mathrm{color}}(p))|$. Three compatibility conditions (F1)–(F3) must all hold: (F1) $r(p) = \tau(p)$; (F2) $\mathfrak{su}(3)$ present; (F3) $\mathfrak{u}(1)$ present. At $p = 3$: $r(3) = 2, \tau(3) = 3$; (F1) fails. At $p = 5$: $F(5) = 11 \ne 5$; (F3) fails. At $p = 7$: partition $7 = 3+2+1+1$ gives $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, $r(7) = 4 = \tau(7)$, $F(7) = 7 = p$; all conditions satisfied. At $p \ge 11$: avoidance-cycle exclusion gives $\tau(p) \le 5$ but $r(p) = p - 2 \ge 9$; (F1) fails. Hence $K_0 = 7$ is the unique prime fixed point of F. (Unconditional; source: dream_paper_v38.pdf §§6.1–6.3.) □

REMARK 3.21 ([R]). The primality clause follows from Axiom (A2) alone. The uniqueness clause ($K_0 = 7$) uses the Weyl structure identity and the fixed-point condition $F(p) = p$; it requires no branch-specific hypotheses. This is a strictly stronger statement than the earlier form.

4. Refinement and Stabilization

Let $\mathcal{T}_0 \subseteq \mathcal{T}_1 \subseteq \mathcal{T}_2 \subseteq \cdots$ be a nested sequence of licensed test families. Each stage induces an observational equivalence $\sim_n$ and quotient $\Sigma_n = \Omega/{\sim_n}$.

Definition 4.1 ([D]). A *refinement chain* on Ω is a nested sequence of licensed test families $(\mathcal{T}_n)_{n\geq 0}$ such that each induced quotient Σ_n is finite.

Proposition 4.2 ([U]). (Monotonicity.) *If $\mathcal{T}_n \subseteq \mathcal{T}_{n+1}$, then $\omega \sim_{n+1} \omega'$ implies $\omega \sim_n \omega'$. Equivalently, every class of Σ_{n+1} is contained in a class of Σ_n.*

Proof. If two configurations are observationally indistinguishable by the larger test family $\mathcal{T}_{n+1}$, they are indistinguishable by every test in the smaller family $\mathcal{T}_n$. Hence every equivalence class for $\sim_{n+1}$ lies inside an equivalence class for $\sim_n$. □

Remark 4.3 ([R]). Proposition 4.2 means refinement proceeds only by partition splitting: classes may stay the same or split into smaller classes, but they never merge. This monotone splitting is the structural engine behind finite stabilization.

Definition 4.4 ([D]). A refinement chain *stabilizes at stage* N if $\Sigma_n = \Sigma_N$ for all $n \geq N$.

Theorem 4.5 ([U]). (Finite Stabilization Theorem.) *Every refinement chain on a finite configuration space Ω stabilizes after finitely many steps.*

Proof. Each quotient Σ_n corresponds to a partition of the finite set Ω. By Proposition 4.2, these partitions are successively finer: no two classes merge. The total number of distinct partitions of a finite set is finite (bounded by the Bell number $B_{|\Omega|}$). Therefore a strictly refining sequence of partitions cannot continue indefinitely: once every possible splitting has occurred, the sequence must stabilize. Hence there exists a stage N such that $\Sigma_n = \Sigma_N$ for all $n \geq N$. □

Definition 4.6 ([D]). Fix a stabilization stage N and define the *stabilized observational quotient*

$$\Sigma_\infty \;:=\; \Sigma_N.$$

Remark 4.7 ([R]). The notation Σ_∞ is mnemonic, not ontological. It records that further admissible refinement no longer changes the quotient. It does not assert that a primitive infinite-limit object has been introduced; no such object exists at this stage. The subscript ∞ means *eventual stability*, nothing more.

Corollary 4.8 ([U]). *The stabilized observational quotient Σ_∞ is well-defined and independent of the particular stabilization stage N chosen.*

Proof. By Theorem 4.5, a stabilization stage N exists. If N' is another stabilization stage with $N' \geq N$, then $\Sigma_{N'} = \Sigma_N$ by the definition of stabilization. □

5. Survivor Chains and Defect Bookkeeping

Before Book I ends, it introduces the primitive bookkeeping that later supports Book III's transport language. At this stage, these are purely combinatorial objects with no dynamical or physical interpretation.

DEFINITION 5.1 ([D]). Let $C_n \in \Sigma_n$. A *survivor chain* issuing from C_n is a nested sequence of classes

$$C_n \supseteq C_{n+1} \supseteq C_{n+2} \supseteq \cdots, \quad C_m \in \Sigma_m \text{ for all } m \geq n.$$

A survivor chain records class-level persistence through successive admissible refinements. It is not a token-level identity claim; it is a quotient-class-level persistence claim compatible with compression.

DEFINITION 5.2 ([D]). Let $C_n \in \Sigma_n$. The *continuation defect* of C_n at stage n is the failure of stage-n admissible data to determine a unique descendant class in Σ_{n+1}.

DEFINITION 5.3 ([D]). For $C_n \in \Sigma_n$, define the *one-step defect number*

$$\delta(C_n) \ := \ \#\{C_{n+1} \in \Sigma_{n+1} : C_{n+1} \subseteq C_n\} \ - \ 1.$$

Thus $\delta(C_n) = 0$ exactly when continuation to the next stage is unique.

PROPOSITION 5.4 ([U]). *For every $C_n \in \Sigma_n$, one has $\delta(C_n) \geq 0$, with equality if and only if C_n has a unique descendant in Σ_{n+1}.*

PROOF. Every element of C_n belongs to some class of the finer partition Σ_{n+1}, so at least one descendant exists. Therefore the number of descendants is at least 1, hence $\delta(C_n) \geq 0$. Equality holds exactly when there is precisely one descendant. □

DEFINITION 5.5 ([D]). For a survivor chain $C_n \supseteq C_{n+1} \supseteq \cdots \supseteq C_m$, its *defect ledger* is

$$\Delta(C_n \rightsquigarrow C_m) \ := \ \sum_{k=n}^{m-1} \delta(C_k).$$

This records the cumulative branching burden along the chain.

PROPOSITION 5.6 ([U]). (Ledger additivity.) *For any intermediate stage ℓ with $n \leq \ell \leq m$,*

$$\Delta(C_n \rightsquigarrow C_m) \ = \ \Delta(C_n \rightsquigarrow C_\ell) + \Delta(C_\ell \rightsquigarrow C_m).$$

PROOF. Immediate from the definition of Δ as a finite sum of one-step defect numbers: splitting the sum at index $\ell - 1$ gives the two partial sums on the right. □

DEFINITION 5.7 ([D]). The *structural entropy* of a configuration space Ω (with respect to the stabilized observational quotient Σ_∞) is

$$S(\Omega) \ := \ \log |\Sigma_\infty|.$$

No thermodynamic, probabilistic, or physical interpretation is assumed. $S(\Omega)$ measures the logarithmic number of admissible distinctions the stabilized quotient can sustain: the *distinguishability capacity* of the system.

THEOREM 5.8 ([U]). (Entropy Monotonicity.) *For any right-stable admissible extension $E : \Omega \to \Omega'$ (one that does not create new defect events),*

$$S(E(\Omega)) \ \leq \ S(\Omega).$$

For defect-mediated extensions (those that do create new defects), the defect ledger Δ *records the entropy budget consumed:*

$$S(E(\Omega)) \;\leq\; S(\Omega) + \Delta(\textit{chain through } E).$$

Proof. A right-stable extension maps each equivalence class of Σ_∞ into at most one equivalence class of the refined quotient, so the number of classes cannot increase: $|\Sigma'_\infty| \leq |\Sigma_\infty|$, hence $S(E(\Omega)) \leq S(\Omega)$.

For a defect-mediated extension, each defect event records exactly one class-splitting event. The cumulative defect Δ bounds the total number of class splits; hence $|\Sigma'_\infty| \leq |\Sigma_\infty| \cdot 2^{\Delta}$ and $S(E(\Omega)) \leq S(\Omega) + \Delta \log 2 \leq S(\Omega) + \Delta$. □

Remark 5.9 ([R]). Theorem 5.8 has two faces: right-stable extensions cannot increase entropy (the quotient only coarsens or stays put), while defect-mediated extensions may increase entropy but only within bounds recorded by the defect ledger. The defect ledger is therefore the complete bookkeeper of entropic budget in the program.

Theorem 5.10 ([U]). (Ledger–Entropy Duality.) *For any survivor chain* $C_0 \supseteq C_1 \supseteq \cdots \supseteq C_n$*,*

$$S(C_0) - S(C_n) \;=\; \Delta(C_0 \rightsquigarrow C_n).$$

Proof. Each step: $S(C_k) - S(C_{k+1}) = \delta(C_k)$ by Theorem 5.8. Summing and applying Proposition 5.6 gives the result. □

Remark 5.11 ([R]). Definition 5.7 (running balance) and Definition 5.5 (transaction log) are two readings of the same object.

Scholium 5.12 ([U]). (Irreversible Preorder from Defect Ledger.) Define a preorder on the history space $\mathrm{Hist}(R)$ by

$$h \preceq h' \iff \Delta(h) \leq \Delta(h'),$$

where $\Delta(h)$ is the cumulative defect ledger weight along history h. Then $(\mathrm{Hist}(R), \preceq)$ is a preorder:

(i) *Reflexivity and transitivity.* Immediate from $\leq$ on $\mathbb{N}$.
(ii) *Forward-only.* Δ is non-decreasing along any admissible continuation, since each transition appends a non-negative one-step defect $\delta(C_n) \geq 0$ (Proposition 5.4). No admissible continuation can decrease Δ.

The quotient $\mathrm{Hist}(R)/\equiv$ (where $h \equiv h'$ iff $\Delta(h) = \Delta(h')$) carries a partial order with antisymmetry by definition.

This is the unconditional arm of the temporal preorder. The conditional arm — interpreting $\preceq$ as a physical temporal order requiring Phase-A stability — is a theorem of Book II (Book II, Scholium II.9.x). No metric, rate, simultaneity structure, or causal propagation speed is derived here.

Corollary 5.13 ([U]). (Holographic Sparsity.) [Conditional on WL-1 stabilization package and LS-2 (imported from Book II).] *For a finite connected region* X *with* m *pairwise* Δ_{sep}*-separated fork sites,*

$$m \log 2 \;\leq\; \log W(X) \;\leq\; (\log K_0)\, |\partial X|,$$

where $W(X)$ is the number of distinct witness outcomes on X and ∂X is the boundary of X. The upper bound is the interface capacity inequality IC-1 (Book II); the lower bound is the fork entropy corollary. The area-law character is a structural consequence of finite alphabet and LS-2, not an imported physical principle.

Proof. Upper bound: by IC-1, each boundary site can contribute at most $\log K_0$ bits of distinct witnessable outcome; with $|\partial X|$ boundary sites the bound follows. Lower bound: the m fork sites are pairwise separated, so each contributes an independent binary choice to $W(X)$, yielding $W(X) \geq 2^m$. □

Remark 5.14 ([R]). Corollary 5.13 is conditional on LS-2, a hypothesis of Book II. It is stated here for architectural completeness because its upper bound (IC-1) is a direct consequence of the primitive finite-alphabet constraint $K_0 < \infty$ established in this book. Full proof requires the collar machinery of Book II.

Scholium 5.15 ([D]). (Canonical Ledger Quotient on G_9.) [Computational certification. Analytic proof open.] Let $G_9 = K_3$-b_3-K_3 be the barbell graph on 9 vertices with $V \cong \mathbb{F}_2^4$ as the WL-1 color space (4 bits: midpoint, connectors, bridge ends, clique bulk). Among the 28 radius-2 single-edge extensions from the base configuration:

(1) 22 preserve all 14 nontrivial orbit-constant refinements (defect-free).
(2) 6 break exactly 8 of the 14 refinements (bimodal obstruction).
(3) All 6 obstructing extensions share the same survivor subspace $W \subset V$ with $|W| = 8$.
(4) The unique linear constraint is $W = \{x \in V : \mathrm{bit}_2 + \mathrm{bit}_4 \equiv 0 \pmod 2\}$.
(5) The annihilator $a = (0, 1, 0, 1) \in V^*$ is the unique codimension-1 kernel direction (the *canonical ledger axis*).

Source: v7.35 Theorem 55.1 (exhaustive BFS enumeration).

Scholium 5.16 ([D]). (Depth-2 Context Ecology on G_9.) [Computational certification.] Under BFS to depth 2 (495 states, 852 transitions) from G_9, the reachable configurations partition into 8 distinct survivor contexts stratified by dimension as $1 + 3 + 3 + 1$:

(1) Dimension 1: 1 context (10 states).
(2) Dimension 2: 3 contexts (16, 14, 10 states).
(3) Dimension 3: 3 contexts (109, 6, 2 states) — the *three-family structure*.
(4) Dimension 4: 1 context (328 states, full V).

The dimension-3 triple provides computational evidence that three distinct survivor families coexist under the admissible dynamics at this substrate scale. Source: v7.35 Theorem 56.1.

Scholium 5.17 ([D]). (Certified Gating Depths on G_9.) [Computational certification.] On G_9 with $V \cong \mathbb{F}_2^4$, the six weight-2 pair-identification hyperplanes have certified gating depths:

Constraint	Gating depth
$\mathrm{bit}_2 + \mathrm{bit}_4 = 0$	1
$\mathrm{bit}_3 + \mathrm{bit}_4 = 0$	2
$\mathrm{bit}_1 + \mathrm{bit}_3 = 0$	2
$\mathrm{bit}_1 + \mathrm{bit}_2 = 0$	2

The depth-1 gate ($\mathrm{bit}_2+\mathrm{bit}_4 = 0$) is the canonical ledger axis identified in Scholium 5.15. Source: v7.35 Proposition 57.3.

Definition 5.18 ([D]). A class $C_n \in \Sigma_n$ is a *stabilizer class* if every survivor chain issuing from C_n has zero defect from stage n onward.

Proposition 5.19 ([U]). *If the refinement chain stabilizes globally at stage N, then every class in Σ_N is a stabilizer class.*

Proof. After stage N, the partitions do not change, so every class has exactly one descendant at every later stage: itself. Hence every later one-step defect vanishes. $\square$

Remark 5.20 ([R]). At the level of Book I, the defect ledger is purely combinatorial bookkeeping. The concepts of continuation, persistence across contexts, transport obstruction, and finite balance will not acquire their full force until Book III. The purpose of introducing the defect ledger here is to fix its arithmetic substrate cleanly at the primitive level, before any dynamical or comparative structure is available. Later books may invoke these objects through their canonical names without returning to the primitive setup.

6. Governing Theorem of Book I

The content of Book I can now be compressed into its governing theorem.

Theorem 6.1 ([U]). (Observational Quotient Theorem.) *Finite admissible-test quotients define a coherent observational state space, and every finite refinement chain stabilizes after finitely many steps.*

More precisely:

(a) *The observational quotient Σ_n is well-defined at every stage and canonical observational content is determined by admissible test equivalence, not by raw token identity.*
(b) *Every refinement chain on a finite configuration space stabilizes: there exists N such that $\Sigma_n = \Sigma_N =: \Sigma_\infty$ for all $n \geq N$.*

Proof. Part (a) is Corollary 3.14. Part (b) is Theorem 4.5. $\square$

Corollary 6.2 ([U]). *No later canonical theorem in the spine may depend on presentation-level distinctions that are invisible to the stabilized observational quotient.*

Proof. By Theorem 6.1(a), the only distinctions with canonical force are those detectable by admissible tests. Presentation-level distinctions collapsed by $\sim_\infty$ carry no admissible witness, and therefore no canonical force. $\square$

Corollary 6.3 ([U]). *The defect ledger is a well-defined combinatorial invariant of survivor chains, additive along concatenation, and vanishes on every survivor chain after the global stabilization stage.*

Proof. Propositions 5.4, 5.6, and 5.19. □

Scholium 6.4 ([R]). Book I establishes only the primitive theorem-bearing kernel: admissible tests, observational quotient, finite stabilization, and defect bookkeeping. It does not prove local stability of observable outcomes, boundary determinacy, source universality, branch legitimacy, or continuum correspondence. Every one of those claims belongs to a later book and may not be silently imported here.

7. Boundary of Book I

The following is an explicit statement of what Book I has and has not proved.

What Book I proves.

(1) Every finite configuration space admits a coherent observational quotient induced by any licensed test family.
(2) Observational equivalence is an equivalence relation; the quotient map is many-to-one precisely where distinct configurations are observationally indistinguishable, forcing singular projection interfaces.
(3) Every finite refinement chain stabilizes after finitely many steps. The stabilized quotient Σ_∞ is well-defined.
(4) The defect ledger is a well-defined additive combinatorial invariant of survivor chains; it vanishes after global stabilization.
(5) The six collapse gates (CG-1 through CG-6) are the pre-conditions for any persistent admissible object.
(6) Behavioral congruence $\sim_n$ is the canonical finest admissibility-consistent equivalence; the maximal admissible class $E_{\max}$ is unique (Thm 3.17).
(7) The collar alphabet size K_0 is prime (Thm 3.20).
(8) Structural entropy $S(\Omega) = \log|\Sigma_\infty|$ is well-defined and monotone non-increasing for right-stable extensions (Thm 5.8).
(9) The defect ledger defines an irreversible preorder on continuation histories (Scholium 5.12).

What Book I does not prove, and may not be assumed here.

(1) *Locality.* No claim is made that observable outcomes in a region are controlled by boundary data. That requires the LS-2 package of Book II.
(2) *Boundary capacity.* No counting bounds on outcome multiplicity are derived here.
(3) *Source universality.* The existence of a universal source object is a theorem of Book IV, not of Book I.
(4) *Persistence and transport.* The transport and persistence machinery of Book III does not exist here. The defect ledger is primitive bookkeeping, not a transport law.
(5) *Continuum structure.* No continuum, differential, integral, or smooth language is licensed at this stage.

(6) *Physical interpretation.* No identification with spacetime, gauge fields, matter, or any physical theory is claimed or implied.

8. Handoff to Book II

Book II begins when the stabilized observational structure established here is localized. The natural next question — once one has a stabilized observational quotient Σ_∞ — is whether the observable outcome on a finite subregion is determined by what happens at its boundary. That question requires introducing regions, collars, and stabilized collar types: all direct descendants of the objects defined in this book.

The transition from Book I to Book II is:

$$\text{adm. process class} \to \text{obs. quotient} \to \text{finite stabilization}$$
$$\to \text{stabilized local types and boundary law.}$$

Book I supplies the first three terms of this chain. Book II supplies the fourth.

9. Dependency Ledger

The following table records the load-bearing items of Book I, their status, and their export destinations.

Theorem	**St.**	**Depends on**	**Exports to**
3.8 Obs. equiv. rel.	[U]	Def. 3.7	Defs. 3.9, 6.1
3.12 Singular interfaces	[U]	3.8, 3.11	Rmk. 3.13; Book V
3.14 Quotient well-def.	[U]	3.8	6.1; Books II–VII
4.2 Monotonicity	[U]	4.1	4.5
4.5 Finite Stabilization	[U]	4.2, fin. of Ω	6.1; Books II–VII
4.8 Σ_∞ well-def.	[U]	4.5	Books II–VII
5.4 $\delta(C_n) \geq 0$	[U]	5.3	5.6, 6.1
5.6 Ledger additivity	[U]	5.5	6.3; Book III
5.19 Stabilizer classes	[U]	4.5, 5.18	6.3; Book III
5.8 Entropy Monotonicity	[U]	5.7, 5.6	5.12; Book III
5.12 Irrev. preorder	[U]	5.4, 5.6	Book II Scholium (Phase-A arm); Book III
5.13 Holographic Sparsity	[C]	3.19; Book II IC-1	Book I boundary; GR Branch Book
6.1 Obs. Quotient Theorem	[U]	3.14, 4.5	Books II–VII; all branch books
6.2 No pres. surplus	[U]	6.1	Books II–VII; Book VII
6.3 Defect ledger	[U]	5.4–5.19	Book III transport machinery
3.15 Behav. Congruence Canon.	[U]	3.14; A1–A4	3.17; Books II–VII
3.17 Admiss.-Class Canon.	[U]	3.15; A1–A4	Books II–VII; Book IV
3.20 K_0 primality	[U]	A2; 3.19	Book II collar machinery; YM Branch Book

Remark 9.1 ([R]). Every unconditional result in this ledger depends only on the standing definitions and the finiteness of Ω. No hypothesis beyond the finite relational framework is invoked. The one conditional result, Corollary 5.13 (Holographic Sparsity), imports LS-2 from Book II; it is recorded here for architectural completeness

only. This unconditionality is the source of Book I's load-bearing character: everything downstream in the spine rests on a foundation that requires no further assumptions to stand.